

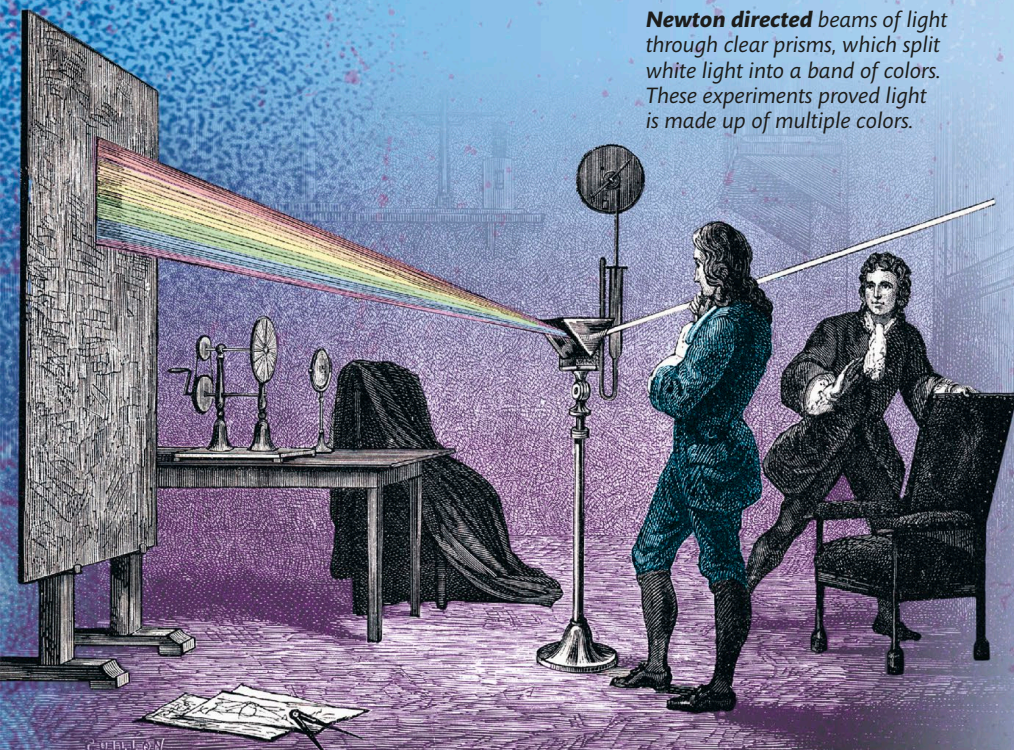
British physicist and mathematician Isaac Newton's laws of motion and law of universal gravitation fundamentally changed the way that humans understood the world. His groundbreaking work revolutionized scientific thought, making him one of the greatest scientists to have ever lived.

Born in Lincolnshire on Christmas Day in 1642, Isaac Newton was brought up there by his grandmother, and suffered from loneliness and ill health. He later attended Cambridge University, but in 1665, it closed due to the Plague, and Newton returned to his home county to avoid contagion. It was here that, reputedly, his transformative theory of gravity first began to evolve, and from an inauspicious start, Newton would rise to become one of the world's most influential scientists.

Planetary motion

Newton's great theory was sparked by his observation of an apple falling from a tree in his grandmother's orchard. The questions this generated—as to why the apple fell downward and what was controlling that movement—gave rise to his three foundational laws of motion and his law of universal gravitation, which together became the backbone of scientific understanding of how the world works. In the 17th century, the accepted view of the Universe was that it was heliocentric, meaning

Newton directed beams of light through clear prisms, which split white light into a band of colors. These experiments proved light is made up of multiple colors.



MILESTONES

GRAVITY INSPIRATION

Observes an apple falling from a tree and begins to formulate his law of universal gravitation in 1666.

REFLECTOR TELESCOPE

Revolutionizes the design of telescopes by using mirrors to reflect light and clarify image in 1668.

KEY PUBLICATION

Publishes *Principia* in 1687, his seminal work outlining three laws of motion and the law of gravity.

SCIENTIFIC LEADER

Accepts the position of President of the Royal Society in 1703; holds the position for 24 years.

SCIENCE OF LIGHT

Discovers the composition of white light; publishes the results in his 1704 text entitled *Optiks*.